

Modifications of carbonate fracture hydrodynamic properties by CO₂-acidified brine flow

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ABSTRACT Acidic reactive flow in fractures is relevant in subsurface activities such as CO₂ geological storage and hydraulic fracturing. Understanding reaction-induced changes in fracture hydrodynamic properties is essential for predicting subsurface flows such as leakage, injectability, and fluid production. In this study, x-ray computed tomography scans of a fractured carbonate caprock were used to create three dimensional reconstructions of the fracture before and after reaction with CO₂-acidified brine (Ellis et al., 2011, Greenhouse Gases: Sci. Technol., 1:248-260). As expected, mechanical apertures were found to increase substantially, doubling and even tripling in some places. However, the surface geometry evolved in complex ways including ‘comb-tooth’ structures created from preferential dissolution of calcite in transverse sedimentary bands, and the creation of degraded zones, i.e. porous calcite-depleted areas on reacted fracture surfaces. These geometric alterations resulted in increased fracture roughness, as measured by surface Z_2 parameters and fractal dimensions D_f . Computational fluid dynamics

(CFD) simulations were conducted to quantify the changes in hydraulic aperture, fracture transmissivity and permeability. The results show that the effective hydraulic apertures are smaller than the mechanical apertures, and the changes in hydraulic apertures are nonlinear. Overestimation of flow rate by a factor of two or more would be introduced if fracture hydrodynamic properties were based on mechanical apertures, or if hydraulic aperture is assumed to change proportionally with mechanical aperture. The differences can be attributed, in part, to the increase in roughness after reaction, and is likely affected by contiguous transverse sedimentary features. Hydraulic apertures estimated by the 1D statistical model and 2D local cubic law (LCL) model are consistently larger than those calculated from the CFD simulations. In addition, a novel ternary segmentation method was devised to handle the degraded zones, allowing for a bounding analysis of the effects on hydraulic properties. We found that the degraded zones account for less than 15% of the fracture volume, but cover 70% to 80% of the fracture surface. When the degraded zones are treated as part of the fracture, the fracture transmissivities are two to four times larger because the fracture surfaces after reaction are not as rough as they would be if one considers the degraded zone as part of the rock. Therefore, while degraded zones created during geochemical reactions may not significantly increase mechanical aperture, this type of feature cannot be ignored and should be treated with prudence when predicting fracture hydrodynamic properties.

KEYWORDS: microtomography, x-ray computed tomography, CFD simulation, fracture, roughness, hydrodynamic property, transmissivity, permeability, hydraulic aperture, degraded zone, carbonate caprock, calcite dissolution

Introduction

Understanding changes of fracture hydrodynamic properties due to contact with acidic fluids is essential in numerous subsurface activities, such as geological storage of CO₂, shale gas production, geothermal energy extraction and nuclear waste disposal. Fractures are of special importance due to their prevalence¹, and potential impacts on flow and reactive transport². Both natural and induced fractures are widely documented in subsurface systems³⁻⁵; including CO₂ storage sites^{6,7}. Geomechanical studies have demonstrated that during CO₂ injection, fractures can be created, activated and propagate in caprocks because of shear stresses⁸, thermal stresses⁹ and crystallization¹⁰. As fractures serve as conduits for flow, their presence may enhance the risks of CO₂ leakage. Therefore, a comprehensive assessment of caprock integrity requires detailed characterization of fracture hydrodynamic properties and knowledge of their evolution over time. During shale gas and geothermal energy extraction, fractures are often intentionally induced. Their hydrodynamic properties control the production and hence profitability of the fields.

Fractures provide pathways for transport of reactive solutes, resulting in thermodynamic gradients that accelerate water-rock interactions. Geochemical alterations of fracture surfaces and flow-path geometry will ultimately modify hydrodynamic properties of caprocks or production formations. In the context of CO₂ geological storage, both simulation results¹¹ and experiments¹²⁻¹⁴ have shown profound alteration of fracture geometry due to the flow of CO₂-acidified brines. Similarly, acidic fracturing fluids may be used to alter the hydrodynamic properties of fractures created or activated during the exploitation of gas-bearing shale formations. Although shale formations typically have a high content of relatively unreactive clay minerals, they often co-exist with other acid-soluble minerals, e.g. carbonates¹⁵. Furthermore, if

induced fractures are not properly contained within the production formation, but instead propagate into the overlying formations, the hydrodynamic properties of these fractures and their evolution will determine the risks of vertical migration of fracturing fluid and gas, and potentially groundwater pollution.

Reactive transport in fractures and the patterns of dissolution and precipitation are controlled by the flow regime¹⁶⁻¹⁸, initial geometry^{2,18}, confining stress¹⁷, mineralogy and brine chemistry^{13,14}. Numerous simulation studies have investigated the evolution of fracture aperture and hydrodynamic properties that result from the interactions between fracture surfaces and reactive flow¹⁸⁻²³. In addition, imaging techniques such as light transmission²⁴ and 3D micro X-ray computed tomography (xCT)²⁵⁻²⁸ provide non-invasive geometric measurement tools, and their applications have enabled observations of acid-driven geometric evolution of fractures^{16,17,25,29,30}. Detwiler¹⁷ observed relatively uniform dissolution at low Damkohler number and channelization at high Damkohler number. This same research group also reported that low Peclet number favors formation of channels, while high Peclet number leads to smoothing of small scale roughness without channelization¹⁶. Furthermore, Szymczak and Ladd²³ found that channelization occurs above a roughness threshold and is favored under the conditions of high reaction rates and intermediate Peclet number. While these studies have greatly advanced our knowledge of some fundamental processes underlying geochemically-driven fracture evolution, they often assumed homogeneous mineralogy and used model fractures.

The effects of mineral spatial heterogeneity on fracture evolution are poorly understood, and need consideration as mineral heterogeneity may result in very complex geometric modifications. In a high-pressure experiment, Ellis et al.¹³ flowed CO₂-acidified brine through an artificially fractured carbonate core from the Amherstburg formation of the Michigan

sedimentary basin. Post-experiment scanning electron microscopy (SEM) imaging revealed extensive dissolution and complex geometric alterations of the fracture created by preferential dissolution patterns, as shown in Figure 1. For example, where the relatively soluble mineral calcite is banded with less soluble dolomite, preferential dissolution of calcite created ‘comb-tooth’ shaped roughness (Figure 1a). In contrast, where calcite is homogeneously mixed with dolomite or non-reactive clay minerals and quartz, very porous calcite-depleted areas were formed, which the authors refer to as “degraded zones” (Figure 1b).

It remains an open question as to whether extensive fracture volume increases resulting from dissolution will manifest as substantial alterations to fracture hydrodynamic properties, and thereby compromise caprock sealing capacity. It is possible that geometric alterations such as increased roughness will counteract the impact of fracture volume increase. In an experiment flowing acidic water through a fractured argillaceous limestone rock sample, Noiriel et al.³⁰ recorded a decrease in permeability that they attributed partly to the increase of fracture roughness.

A large volume of literature exists on fracture roughness and its impacts on flow. One-dimensional models based on the distribution of local apertures have been proposed to incorporate roughness by using a weighted average or standard deviation of apertures³¹⁻³³. Another modeling approach widely adopted is the Local Cubic Law (LCL)^{18,19,34-37}. It assumes that the cubic law for smooth parallel wall fractures holds locally, and solves Reynolds equation locally. However, these models do not provide accurate estimations of fracture hydrodynamic properties at relatively high, yet realistic roughness^{29,38,39}.

Computational fluid dynamics (CFD) simulations have been shown to provide accurate calculations for flow in fractures across a range of plausible fracture roughness³⁸⁻⁴⁰. By using

different meshing tools and applying different degrees of smoothing, Crandall et al.³⁹ generated six fracture meshes with different degrees of roughness. They found that increase in fracture roughness results in more tortuous flow and lower transmissivity. Yet, roughness in these studies has typically been decoupled from the spatial distributions of minerals and the resulting dissolution patterns.

In this study, our objective was to investigate the relationships between the change in fracture surface geometry and hydrodynamic properties resulting from acid-promoted dissolution. In particular, we investigated the effects of significant fracture volume increase and roughness increase that arises because of mineral spatial heterogeneities. Based on the outcome of the experiment by Ellis et al.¹³, xCT scans of the fractured core were used to reconstruct the fracture geometries before and after reaction with flowing CO₂-acidified brines. CFD simulations were performed on 3D finite-element meshes generated from the fracture geometries, allowing a comparison of predicted hydrodynamic aperture with mechanical aperture. To determine if simpler modeling approaches can capture fracture surface geometry effects, hydrodynamic properties were also estimated using a one-dimensional statistical model and a two-dimensional LCL model. The basis for comparison of these model predictions are the fracture hydrodynamic properties calculated from the CFD simulation results. It was not possible to make comparisons with experimental permeabilities, which were not measurable under the flow condition of the experiment, as explained by Ellis et al.¹³.

Our study is novel in two aspects. First, the fracture geometries used are the actual results from reaction of a caprock specimen, which is a typical dolomitic limestone that is rich in mineral heterogeneity. Second, we developed a novel segmentation approach to examine the effect of degraded zones, i.e. the calcite-depleted porous rock on reacted fracture surfaces.

Theoretical Basis

Fracture transmissivity, T , and permeability, k , are both measurements of the ease for fluid to flow through a fracture, as described by the relationship between volumetric flow rate, Q , and the pressure gradient, ∇P , through Darcy's law:

$$Q = -T \frac{(\nabla P)}{\mu} = -\frac{Ak}{\mu} \nabla P \quad (1)$$

where μ is the fluid viscosity and A is the cross-sectional area of the fracture perpendicular to flow. Theory for single phase flow in a smooth parallel fracture⁴¹, frequently referred to as the cubic law, states that T is related to the cube of mechanical aperture, b , the vertical distance between two fracture walls. When the assumption of smooth parallel walls breaks down, the aperture is replaced with its effective value, the hydraulic aperture b_h ³³,

$$T = -\frac{wb_h^3}{12} \quad (2)$$

$$k = \frac{b_h^2}{12} \quad (3)$$

where w is the width of the fracture. Properties T , k and b_h provide three different ways to characterize fracture hydrodynamic properties, and they are related as shown in eqns 1, 2 and 3. The value of the fracture T and fracture intrinsic k is that they can be directly compared with field measurements, lending insights on how the fracture may impact the hydrodynamic properties of the formation. The b_h can be readily compared with b , which can be derived from image analyses or estimated from fracture volume, to investigate whether and how the flow behaves differently from flow in a parallel wall fracture.

Methods

Segmentation and Meshing

The xCT scans used in this study had voxel resolution of 27 μm . Two separate areas of the fracture were selected to avoid regions that were not amenable to meshing, including a horizontal fracture and the multi-fractured edges. The selected fracture areas have a length of 2.176cm and 3.078cm for Area-1 and Area-2 respectively, and a width of 1.296cm for both areas (Figure 2). xCT images corresponding to the selected fracture areas were processed to isolate the fractures and rock matrix immediately surrounding the fracture boundary on the grayscale xCT images. To segment the grayscale xCT images, we applied Otsu's method⁴², which involves searching for thresholds that minimize interclass variance. As an example, Figure 3a and 3c shows the grayscale xCT image for a cross-sectional slice before the reaction. Figure 3b is the histogram for the boxed area of figure 3a showing the threshold generated by Otsu's method. The resulting binary segmentation separating the fracture from the rock is shown in Figure 3d.

The after-reaction xCT images did not show a clean interface, but instead showed fuzzy zones between the rock and the fracture (Figure 3g). These areas have been previously identified using SEM images of core sections (Figure 1b) as 'degraded zones'¹³, porous areas created by preferential dissolution of calcite from regions where it is intermixed with less-reactive minerals. To examine the effect of depleted zones, a ternary segmentation method was devised to classify voxels as either fracture, rock, or degraded zone (Figure 3h). This method did not simply segment voxels based on grayscale intensity, which would have overestimated the degraded zone due to the ubiquitous presence of transitional grayscale intensities at all edges. Instead, an algorithm was developed in which a window with a specific size moves along the rock-fracture interface to identify zones with intermediate grayscale intensities, and only zones with thickness above a cutoff value was segmented as 'degraded zone'. A wide range of window sizes and

cutoff thicknesses were tested. Numerous visual inspections were conducted to compare the ternary-segmented images with the original xCT images to ensure that the algorithm was accurately characterizing the degraded zone.

To examine the impact of the degraded zones on flow, a bounding analysis was conducted to constrain reality between two extremes. In one scenario, the degraded zones were treated as rock (Figure 3i), assuming the degraded zones are impermeable. In the other scenario, the degraded zones were treated as fracture (Figure 3j), assuming no resistance to flow. Reality falls somewhere between these two scenarios. Comparing flow in the two scenarios provide some insights on the impacts of the degraded zone on fracture hydrodynamic properties.

The segmented images were imported into a 3D volumetric mesh generator iso2mesh⁴³, producing meshes such as that shown in Figure 4. Given the complexity of the fracture geometry, unstructured tetrahedral meshes were generated. Minimal smoothing and simplifications were applied to preserve the complexity of the fracture geometries. Meshing parameters were optimized to ensure good mesh quality (e.g. orthogonality and skewness) and numerical performance (e.g. convergence and computational cost). Six 3D meshes were generated including three meshes each for Area-1 and Area-2: before-reaction binary segmentation, after-reaction with degraded zone treated as rock, and after-reaction with degraded zone treated as fracture. Each mesh contains on the order of 10^6 grids following optimization.

To quantify uncertainties deriving from the choice of mesh density, a smaller area of Area-2 was analyzed using a range of mesh densities. Specifically, the selected Area-2 fracture was the after reaction one with degraded zones treated as rock. This test area was regarded to be the best case for the study of sensitivity to mesh density for two reasons. First, its geometry is the most difficult to mesh, which requires finer grids to capture sharp features. Second, the apertures of

the test area cover a wide range of the aperture values of the whole area, ranging from 0 to 1998 μm . Meshes of the test area were generated using a spectrum of meshing parameters, resulting in density increases by a factor of 2 to 20. CFD simulations of these test meshes showed that higher mesh densities result in slightly lower values of hydraulic aperture. An increase of the mesh density by more than 20 times leads to a decrease of 28 μm . Therefore, we can assume that the uncertainty associated with mesh densities is approximately 28 μm .

Geometric characterizations

To statistically characterize fracture surface geometries, each mesh was converted into a 3D volumetric image with the same voxel resolution of 27 microns as the original xCT images. Fracture aperture was measured as the total number of fracture pixels in each column spanning the two fracture walls in the 3D volumetric image.

Surface roughness was characterized using surface Z_2 parameter and fractal dimension D_f . For a 3D surface with resolution of Δx and Δy on the x- and y- direction respectively, the surface Z_2 parameter is calculated using equation (4)⁴⁴,

$$Z_2 = \left\{ \frac{1}{(N_x-1)(N_y-1)} \left[\frac{1}{\Delta x^2} \sum_{i=1}^{N_x-1} \sum_{j=1}^{N_y-1} \frac{(z_{i+1,j+1}-z_{i,j+1})^2 + (z_{i+1,j}-z_{i,j})^2}{2} + \frac{1}{\Delta y^2} \sum_{j=1}^{N_y-1} \sum_{i=1}^{N_x-1} \frac{(z_{i+1,j+1}-z_{i+1,j})^2 + (z_{i,j+1}-z_{i,j})^2}{2} \right] \right\}^{\frac{1}{2}} \quad (4)$$

where N_x and N_y are the numbers of discrete grids along the x- and y- axis, and z is the height of the surface. A large surface Z_2 parameter corresponds to large variation of the local surface slope and thus corresponds to higher roughness.

Fractal dimensions, D_f , for 2D cross-sections along the flow direction (y) were calculated using the bandwidth method^{39,45}. Standard deviation of the vertical displacement, σ_s , was calculated using equation (5) for a range of window size, s .

$$\sigma_s = \sqrt{\frac{1}{N_y - s} \sum_{i=1}^{N_y - s} (z_{i+s} - z_i)^2} \quad (5)$$

The parameter s ranges from Δy , and is cut off at the long wave length, which is approximately one sixteenth of the length of fracture⁴⁶. A linear fitting was applied on the logarithms of σ_s and s to obtain the Hurst exponent, H . Fractal dimension, D_f , is then calculated from H using equation (7).

$$\sigma_s \propto s^H \quad (0 < s < L/16) \quad (6)$$

$$D_f = 2 - H \quad (7)$$

For each fracture surface, calculations of surface Z_2 parameter and D_f used a resolution (Δx and Δy) of 27 μm . The number of grids along flow direction (y) is 806 and 1140 for Area-1 and Area-2, respectively, and the number of grids across the flow direction (x) is 480 for both areas. Therefore there are 480 cross-sections along the flow direction for each surface. D_f was calculated for each cross-section and the average is used for the surface. For each fracture geometry, surface Z_2 parameters and D_f were calculated for both the upper and lower fracture surfaces (which were assigned arbitrarily, as neither surface is up or down). Since the flow is affected not only by the surface roughness, but also by the degrees of matching between the upper and lower surfaces⁴⁷, a midsurface⁴⁸ defined as the mean of the two fracture surfaces was used, providing a representation of the fracture undulation. Roughness of the midsurface was also characterized for all the fracture geometries.

CFD Simulations

CFD simulations were conducted by importing the finite-element tetrahedral meshes into OpenFOAM (OpenCFD Ltd, ESI Group) and using its tools to numerically solve the continuity and momentum equations. A steady state solver (simpleFoam) was used. Momentum was solved using the smoothSolver, and pressure was solved with the Geometric-algebraic multi-grid (GAMG) solver. The momentum and continuity equations were coupled using the SIMPLE (Semi-Implicit Method for Pressure-Linked Equations) algorithm. A linear scheme was used to interpolate values from cell center to face center, and a limited non-orthogonal correction with $\psi = 0.7$ was selected to evaluate the surface normal gradient from gradient of values at the cell centers. The convergence criterion for steady state was set as when the normalized residual in the solution falls below 10^{-5} .

Boundary conditions used in the simulations are no-slip wall, fixed pressure at the outlet, and uniform velocity profile at the inlet. The test velocities varies from 10^{-6} m/s to 10^{-3} m/s, ensuring a flow regime of creeping flow, i.e. $Re < 1$. Reynolds number (Re) was calculated for all fracture geometries, using the equation⁴⁹

$$Re = \frac{4\rho Q}{p\mu} \quad (8)$$

where ρ is fluid density and p is the perimeter of the channel. Perimeters were calculated for all the fracture cross-sections perpendicular to flow direction in every fracture. Reynolds number calculated from the velocity range and perimeter range vary between 2×10^{-4} and 0.44. Fluid properties of water were used in all the simulations, with $\rho = 998.2$ kg/m³ and $\mu = 10^{-3}$ Pa·s. Four different velocities were used for each fracture geometry to capture variability in the calculations, which resulted in negligible variations in the simulation results, and therefore, only averages are reported for each mesh geometry. The CFD simulations resulted in pressure

gradient fields, which together with the flow rate were used to calculate fracture T , k , and b_h using equation (1)-(3).

1D empirical model

For comparison, the one-dimensional empirical model proposed by Zimmerman and Bodvarsson³³ was also used. The model (eqn (9)) estimates b_h from b , and accounts for the impacts of roughness by incorporating standard deviation of mechanical aperture, σ , and contact area ratio, c , the ratio of contact area over the fracture surface area.

$$b_h^3 \approx \bar{b}^3 \left[1 - \frac{1.5\sigma^2}{\bar{b}^2} \right] (1 - 2c) \quad (9)$$

In this study, contact area ratios within Area-1 and Area-2 are negligible.

2D LCL model

Also for comparison, the 2D LCL model³⁵ was used. In this model, the cubic law is assumed applicable locally. The two-dimensional steady state mass balance equation is given as in (eqn(10)).

$$\frac{\partial}{\partial x} \left[b^3(x, y) \frac{\partial P(x, y)}{\partial x} \right] + \frac{\partial}{\partial y} \left[b^3(x, y) \frac{\partial P(x, y)}{\partial y} \right] = 0 \quad (10)$$

Five-point central finite difference numerical approximation was used to solve the equation at each cell, with the harmonic mean of the apertures of the adjacent cells taken as equivalent aperture on the edge. Vertically averaged velocities are then calculated from the pressure fields and Darcy's law (eqn(11) and (12)).

$$\overline{u_x} = \frac{b(x, y)^2}{12\mu} \frac{\Delta P}{\Delta x} \quad (11)$$

$$\overline{u_y} = \frac{b(x, y)^2}{12\mu} \frac{\Delta P}{\Delta y} \quad (12)$$

Results and Discussion

Fracture Geometric Characterization

Table 1 presents the mesh properties of six fracture geometries. The significant fracture volume increase caused by calcite dissolution is reflected in the higher numbers of cells for the four after-reaction meshes (Table 1). The after-reaction meshes also required finer grids at the geometrically sharp locations to accurately represent surface roughness. As demonstrated in Figure 5a-c, e-g, the distributions of mechanical aperture (b) for both Area-1 and Area-2 shift to larger apertures, and span a broader range after reaction. These findings are consistent with the distributions reported in Ellis et al.¹³ for the whole fracture. When degraded zones are treated as part of the rock, the histograms shift less (Figure 5c and 5g) relative to when the degraded zones are instead treated as fracture (Figure 5b and 5f).

Table 2 documents all of the major statistics of the mechanical apertures for the six meshes. After reaction, the average aperture of Area-1 doubled. For Area-2, the average aperture tripled. The maximum aperture recorded for the after-reaction meshes increased much more than the average aperture, leading to larger standard deviations.

Degraded zones account for less than fifteen percent of the fracture volumes, and yet cover 70% and 80% of the fracture surfaces in Area-1 and Area-2, respectively. Comparing the aperture distributions of the two different treatments of degraded zones (Table 2), average aperture differences are 141 μm and 271 μm for Area-1 and Area-2, respectively. The differences are due to the relatively small fracture volume occupied by degraded zones, which is also reflected in the thickness histograms for the degraded zones that are shown in Figure 5d and 5h. Larger standard deviations are observed when degraded zones are considered as fracture.

Table 1 tabulates the roughness parameters. The surface Z_2 parameters are generally larger than the values reported previously⁴⁴, which can be attributed to the fine resolution of the meshes. The D_f values calculated for all the fracture meshes fall in the 1 to 2 range of a true fractal, and fluctuate around 1.5, the value for a Brownian fractal profile⁵⁰, justifying the adoption of fractal dimension and the method used to calculate D_f in the fracture meshes.

The dissolution reactions significantly enhanced the roughness of the fracture surfaces in both Area-1 and Area-2. Both the surface Z_2 parameters and D_f increase after reactions. This trend is consistent for the upper and lower walls, as well as the midsurface of the fracture meshes, implying that the changes of the two fracture surfaces are not perfectly correlated. However, the lower roughness of the midsurface relative to the fracture walls indicates some degree of symmetry and registry between the two fracture walls. If degraded zones are treated as fracture, the increase in roughness relative to the before-reaction meshes is limited; whereas the roughness increase is more significant when degraded zones are treated as rock. The impacts of degraded zones on surface roughness measured by both surface Z_2 parameters and D_f are consistently observed for both areas.

Figure 6 shows the comparison between the changes in aperture (Figure 6a) and the spatial distribution of degraded zones (Figure 6b). The white regions of Figure 6b, which indicate the absence of degraded zones, are generally correlated with areas of great aperture increase shown in Figure 6a. In contrast, presence of thick degraded zones as shown in darker color in Figure 6b coincide with some of the regions where aperture increase is limited. In regions such as the blue band at the bottom of Figure 6a, aperture increase is small and degraded zones are absent or thin, thus indicating bands of unreactive minerals. The large surface coverage of degraded zones and the variable thickness shown within degraded zones (Figure 6b) are consistent with the

observation that treating the degraded zone as rock produces the largest after-reaction increase in fracture roughness.

Fracture hydrodynamic properties

The pressure and velocity fields are shown in Figure 7. Examples of pressure fields are shown only for the before-reaction meshes. The CFD inferred T and k values of the after-reaction meshes are larger than those of the before-reaction meshes (Figure 8), indicating an increase in fracture hydrodynamic properties following exposure of the fracture to CO₂-acidified brine flow. When the degraded zones within Area-1 are treated as fracture, T increased by a factor of 8, from the value before reaction, and k increased by a factor of 4. When the degraded zones in Area-2 are treated as fracture, both T and k increased by more than an order of magnitude. When the degraded zones are treated as rock, however, the margin of the increases in T and k narrowed significantly in both areas. For Area-1, T and k of the mesh for which degraded zones are treated as rocks are approximately half of the values for when degraded zones are treated as fracture. Differences in T and k between the two after-reaction meshes are even larger for Area-2.

The hydraulic aperture inferred from CFD simulations b_{hCFD} are consistently lower than the average b values derived directly from the xCT images (Figure 9). For Area-1, the before-reaction b_{hCFD} is approximately 150 μm less than the b , while the after-reaction differences are twice this more. For Area-2, the initial difference between b_{hCFD} and b is approximately 100 μm , but this difference increased to more than 500 μm after reaction. The discrepancy between b and b_{hCFD} is smaller when degraded zones are treated as fracture. Values of hydraulic aperture estimated using the 1D roughness model developed by Zimmerman and Bodvarsson³³ and the 2D (LCL) model³⁵ are also smaller than b . These values, however, are consistently larger than

the b_{hCFD} . In addition, the differences between hydraulic apertures estimated from the 1D and 2D model and b_{hCFD} and are much larger than the uncertainty in b_{hCFD} associated with increasing mesh density. Finally, although both b_{hCFD} and b increased following reactive flow due to dissolution reactions, the changes in b_{hCFD} are not proportional to the changes in b . The ratios of b_{hCFD} after reaction over b_{hCFD} before reaction are always smaller than the ratios of b , meaning that increase in b_{hCFD} is less than the increase in b .

The Impacts of Fracture Roughness

In a fracture with smooth parallel walls, b and b_h are equivalent, and therefore, b can be used directly to estimate fracture hydrodynamic properties. Monitoring effluent chemistry and using mass balance provide one way to quantify dissolution and infer changes in average b ³⁰. Alterations in b due to reactions can also be calculated from reactive transport modeling^{19,37,51}. In addition, applications of imaging techniques such as μ CT scans^{25,27} and light transmission²⁴ allow characterization of fracture morphology and b from images.

For rough-wall fractures, b_h deviates from b (e.g. Figure 9), and the extent of the deviation is primarily controlled by the degree of roughness. It has been pointed out in previous work^{16,18,22,30} and confirmed in our study that reactions change both the volume and roughness of fracture. Moving from the before-reaction geometry to the after-reaction geometry where the degraded zones are treated as rock, fracture roughness increases and streamlines in the fractures become more tortuous, deviating more from the parallel streamlines that define b and b_h equivalence (Figure 7). As a result, the deviation of b_h from b , as measured by the ratio between b_h and b grows as roughness increases (Figure 10). This variation in the deviation of b_h from b greatly complicates quantification of the changes in hydrodynamics properties caused by reaction. For

example, for the after reaction geometry of Area-2 with degraded zone treated as rock, b has tripled while b_h shows only doubling (Figure 9b). Therefore, if a threefold increase in b_h is assumed based simply on the observed changes in b , k and T are overestimated by more than 100% and 200%, respectively, due to the power law relationships (eqn. (2) - (3)). As k and T are the major inputs for most hydrodynamic models, these overestimations will propagate into the model predictions of key performance parameters such as leakage risks of CO₂ storage and shale gas production.

To account for variations in b along a fracture, Tsang and Witherspoon³¹ used a weighted average of b to estimate T . A subsequent study observed that the ratio of b_h and b is related only to the variance of log-aperture, and hence b_h can be estimated from the distribution of fracture b ³². Based on a comprehensive review, Zimmerman and Bodvarsson³³ proposed a model to account for the impacts of roughness by incorporating standard deviation of b and contact area ratio. In addition to the statistical approach, the 2D LCL model is also widely used. In fractures with low roughness, its capability of capturing fracture hydrodynamic properties was justified and the uncertainties caused by the model were considered negligible compared with other processes³⁷.

However, for the fracture geometries in this study, both the 1D statistical model developed by Zimmerman and Bodvarsson³³ and 2D LCL model that assumes cubic law applies locally largely over-estimate the flow, and failed to fully capture the impacts the roughness. The poor performance of these modeling approaches is primarily rooted in the severe roughness caused by preferential dissolutions of calcite relative to less soluble silicate and dolomite.

Mineral Heterogeneity and Evolution of Fracture Hydrodynamic Properties

Changes of fracture roughness after exposure to acidic flow are primarily controlled by the spatial distribution of reactive minerals and less soluble minerals. One type of sedimentary feature that results in roughness increase is banding of reactive and unreactive minerals. The cross sections of the fracture along the flow direction at nine x values shown in Figure 11 illustrate that though extensive dissolution has resulted in substantial b enlargement; there are still some regions along the flow pathway where aperture increase is limited due to a contiguous transverse stricture, which exists because of a sedimentary layer of relatively insoluble minerals.

Mineral heterogeneity also affects flow by creation of reaction degraded zones. Two core-flooding studies on different carbonate rocks observed the formation of ‘degraded zones’ along the fracture boundary following the flow of acidified brines^{13,30}. The formation of degraded zones results from the dissolution of calcite from within a matrix of relatively insoluble minerals (e.g., dolomite and silicates). Our study demonstrated that the range of possible transmissivity outcomes is large depending on whether or not there is flow through the degraded zones, even though they account for only a relatively small fraction of the fracture volume. The primary reasons are highlighted by the degraded zone occurrence and thickness map shown for Area-2 in Figure 6b, where degraded zones cover a large fraction of the fracture surface and have a large impact on roughness (Table 1). The large impact of degraded zones on fracture roughness is also consistent with previous findings⁵², which documented higher roughness for coated natural fracture surface than uncoated ones.

Our studies on the impacts of geometric alterations on fracture hydrodynamic properties can lend some insights for predicting their evolution as a result of longer exposure to CO₂-acidified brine. For the case of banding of calcite with dolomite, roughness increased dramatically due to preferential dissolution of calcite. However, because the preferential dissolution of calcite is due

to differential solubilities of the two minerals, longer-term contact with CO₂-acidified brine will cause dolomite to also dissolve. Consequently, constraints on flow by the dolomite bands will be reduced, and fracture T and k will increase. Since dolomite dissolves slowly compared with calcite, roughness will keep growing, but the effect of this roughness may diminish as the b increases. Therefore, increase of fracture hydrodynamic properties will be disproportional to the fracture volume increase. If instead, calcite is banded with non-reactive minerals such as clay minerals, the non-reactive bands will persist as constraints on the flow and fracture hydrodynamic properties may change only negligibly in response to preferential dissolution of calcite. However, sharp changes in fracture hydrodynamic properties may happen if the geomechanical strength of the non-reactive bands is compromised.

In rocks in which minerals are mixed more homogeneously, i.e. not in bands parallel to sedimentary bedding, preferential dissolution of calcite leads to formation of the degraded zone. If the fluid in the degraded zone is hydraulically isolated, existence of the degraded zone will serve as a protective layer against further erosion, for ions need to diffuse through the degraded zone to reach the reaction front. If the degraded zones are composed of dolomite, dolomite will start to dissolve given longer exposure to CO₂-acidified brine, and the degraded zones will evolve towards the direction of the extreme scenario when degraded zones are considered as fracture. Dissolution of dolomite may be faster compared to the fracture surface due to increase of reactive surface area created by the degraded zones. In contrast, if the degraded zones are composed of aluminosilicates left after preferential dissolution of calcite, they will be preserved even with longer exposure to reactive fluid, and the constraints on flow will remain. Another possibility in this case is decohesion of the aluminosilicates. As inferred from the experimental observations by Ellis et al.¹⁴ and Noiriel et al.³⁰, the mobilized particles will be transported by the

flow and may be carried through the fracture, which leads to b increase and smoothing of fracture walls, or they may be trapped and clog the flow pathway, which results in reduction in fracture hydrodynamic properties.

Conclusions

The discrepancy between hydraulic aperture, b_h , and mechanical aperture, b , and the incapability of the 1D statistical model and 2D LCL model to capture the differences imply that prudence is required when studying the impacts of reactive transport on fracture hydrodynamic properties. Coupling the detailed fracture geometry reconstructed from xCT images and CFD simulations provides more accurate estimations of fracture hydrodynamic properties. Our study not only has successfully applied this method, but also has extended it to study true geometrical alterations – high roughness and degraded zones, caused by reactive flow. We used xCT images collected before and after a flow-through experiment and CFD simulations to study the changes of fracture hydrodynamic properties due to reactive flow. We found that modifications of hydrodynamic properties are constrained by both the volume dissolved and the development of specific geometric features that can be traced to the underlying mineralogy. In our study, the increase in fracture hydrodynamic properties is mitigated by an increase in roughness caused by zones of banded reactive and non-reactive minerals and the creation of degraded zones. The high degrees of roughness caused by complex mineralogical spatial distributions lead to large discrepancy between b and b_h calculated from CFD simulations. In such cases, the use of b that can be derived from imaging techniques, reactive transport modeling and experimental effluent analyses to estimate fracture hydrodynamic properties and their evolution will result in overestimation of permeability and transmissivity by a factor of two or more. In addition, the 1D

statistical model and the 2D LCL model failed to capture flow accurately and overestimate the hydrodynamic properties relative to the CFD simulations. Therefore, if b_h estimated from the 1D and 2D models are used to estimate hydrodynamic properties, the errors will propagate into subsequent modeling, leading to overestimation of leakage risks or production prediction. More experimental studies correlating mineralogical spatial distribution and geometric alterations are needed in the future, as well as the efforts dedicated to incorporating the impacts of the resultant roughness efficiently. Such studies will benefit parameterization of the large-scale models used to predict leakage of CO₂ from geologic storage reservoirs and gas from hydraulically fractured shale formations.

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Author contributions

The manuscript was written with equal contributions from all authors. All authors have given approval to the final version of the manuscript.

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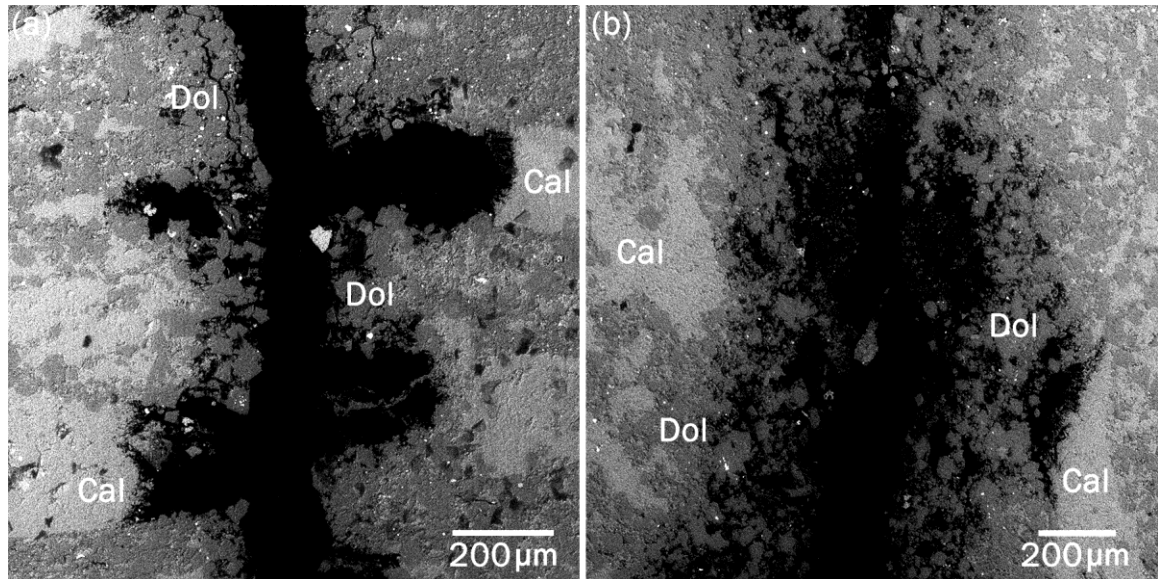
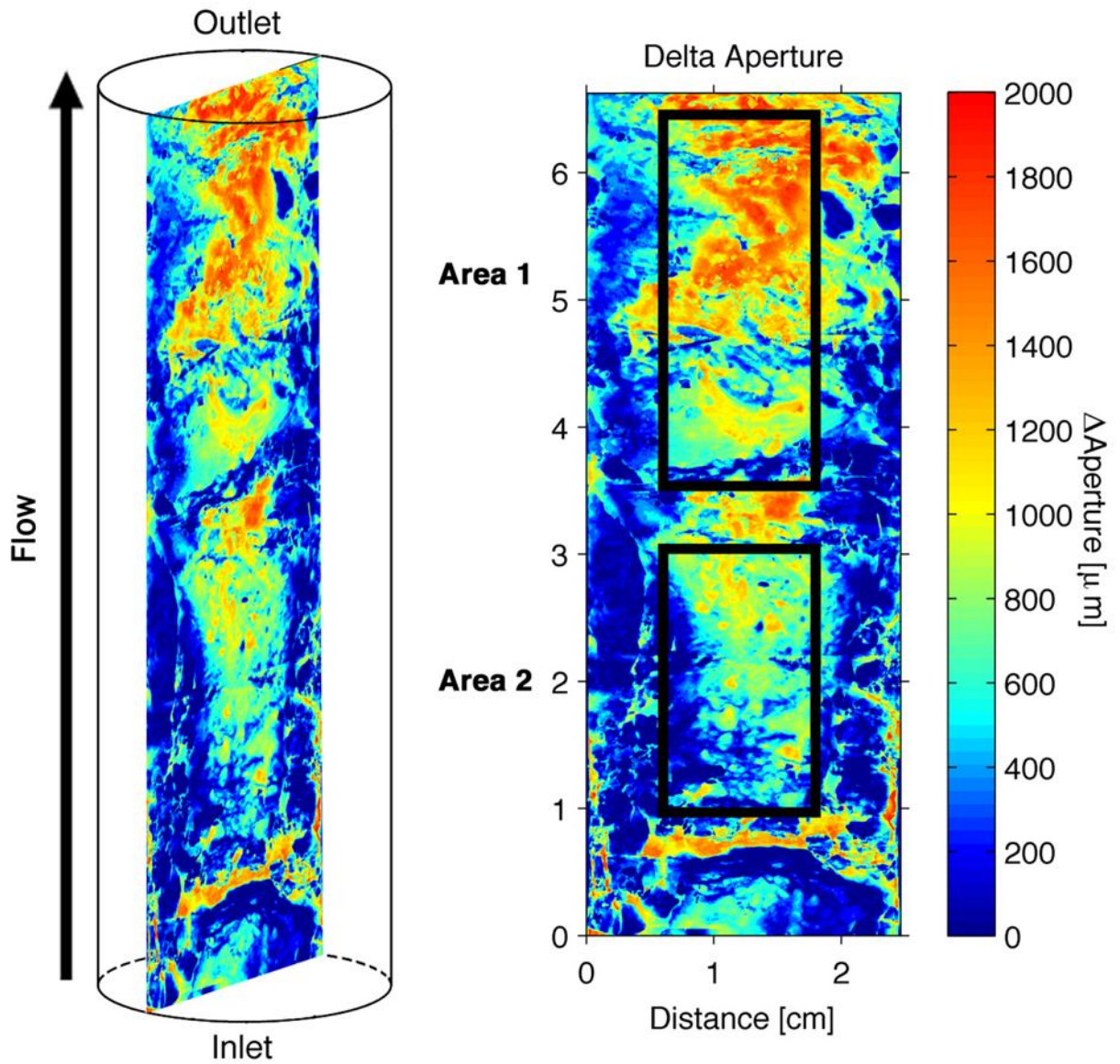


Figure 1. Scanning electron microscopy (SEM) images in back-scattered electron (BSE) mode, showing unique fracture surface geometries that resulted from reaction with CO₂-acidified brine (as a result of the experiment presented in Ellis et al, 2011). (a) comb-tooth wall geometry resulting from receding calcite bands and persistence of less soluble minerals, (b) porous ‘degraded zone’ along the fracture wall created by preferential dissolution of calcite.



647

648 **Figure 2.** Schematic of the fractured core (left) with a map of fracture aperture change due to
 649 reactions with CO₂-acidified brine flow (based on Ellis *et al.*, 2011), with black boxes indicating
 650 two areas selected for CFD simulation in this study (right). (This map is different from the
 651 aperture change maps generated in this work, as a binary segmentation was performed for both
 652 before and after reaction xCT images.)

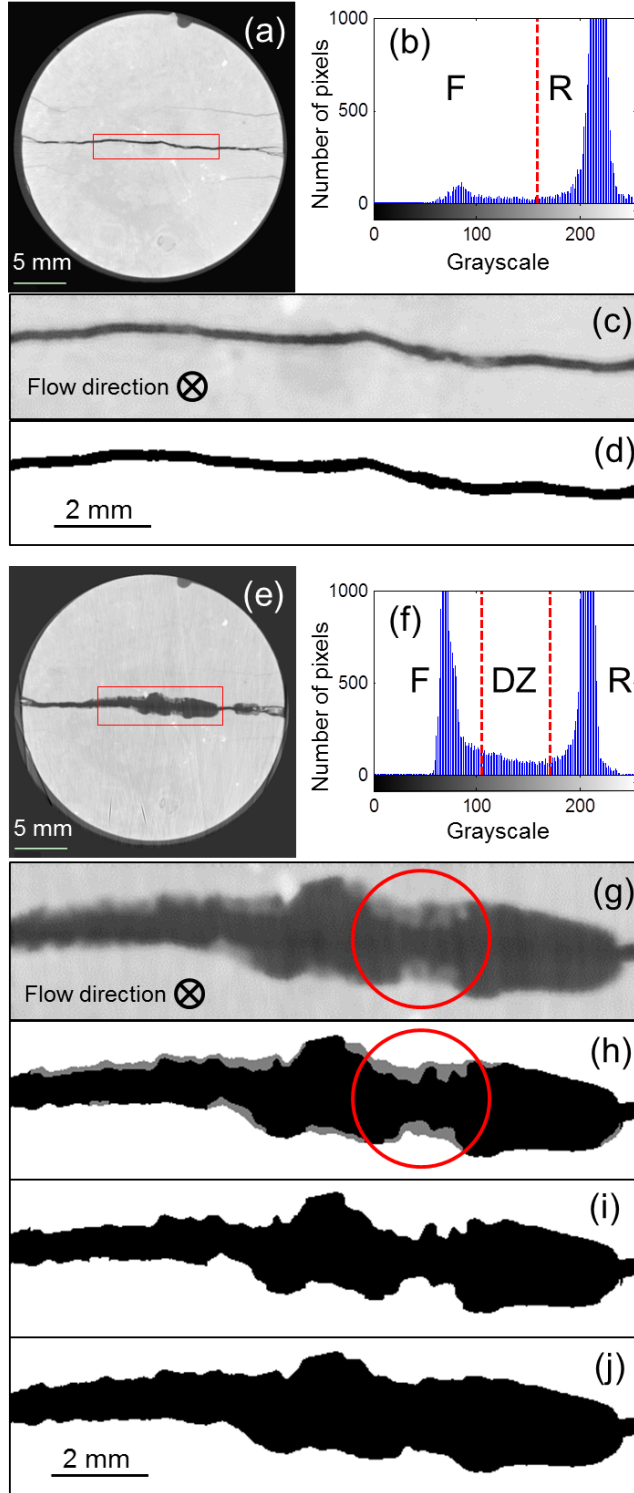
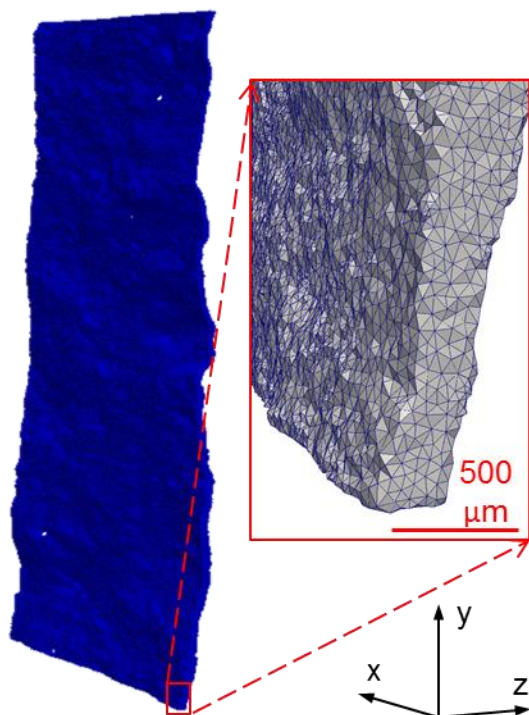


Figure 3. Example images depicting the geometry of the fracture, with a focus on the degraded zone and the ternary segmentation method developed in this work. The location for these images is in Area-2 at $y=5.75$ cm according to the scale shown in figure 2. (a) xCT image of a cross-

657 section of the fracture taken before the experiment, with the study area identified in the red box,
658 (b) histogram for the grayscale image of the study area in (a), F-fracture, R-rock, (c) close-up of
659 the grayscale xCT image of the study area in (a), and (d) segmented image with black
660 corresponding to the fracture and white corresponding to the rock, (e) xCT image of the same
661 fracture cross-section, taken after the experiment, with the study area indicated in the red box, (f)
662 histogram for the grayscale image of the study area in (e), F-fracture, R-rock, DZ-degraded
663 zones, (g) close-up of the grayscale xCT image, with a circle indicating an area where there is a
664 degraded zone, (h) ternary segmented image with degraded zones shown in gray, (i) binary
665 segmented image showing degraded zone treated as rock, and (j) binary segmented image
666 showing degraded zone treated as fracture.

667



668

669 **Figure 4.** Finite-element tetrahedral mesh of Area-2 before reaction, and a close-up of the mesh

670 in the red box.

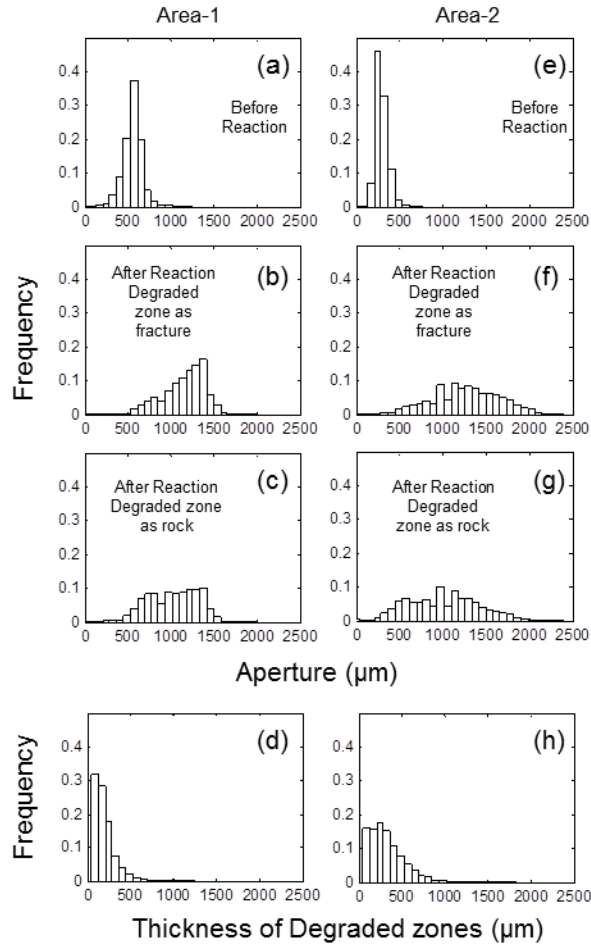


Figure 5. Histograms of the distributions of mechanical apertures for Area-1 (a) before reaction, (b) after reaction with degraded zone treated as fracture, (c) after reaction with degraded zone treated as rock, and Area-2 (e) before reaction, (f) after reaction with degraded zone treated as fracture, (g) after reaction with degraded zone treated as rock. Histograms of the thickness of degraded zones for (d) Area-1, and (h) Area-2.

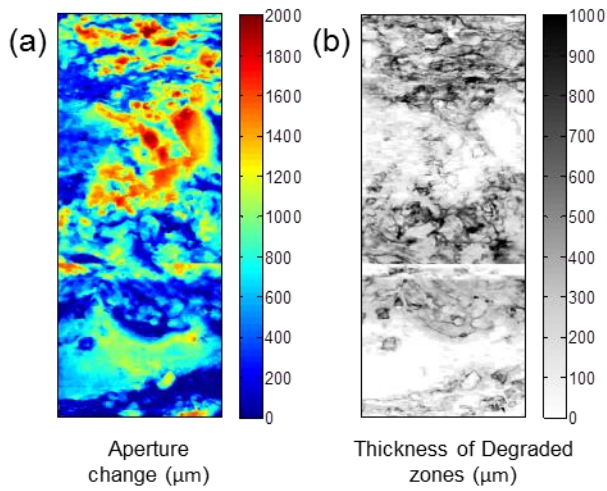


Figure 6. (a) Changes of aperture between after-reaction geometry degraded zones treated as rock and before-reaction geometry of Area-2, and (b) a map of the occurrence and thickness of degraded zones. The linear feature at approximately one third of the area from the bottom is due to the fact that two scans were stitched together.

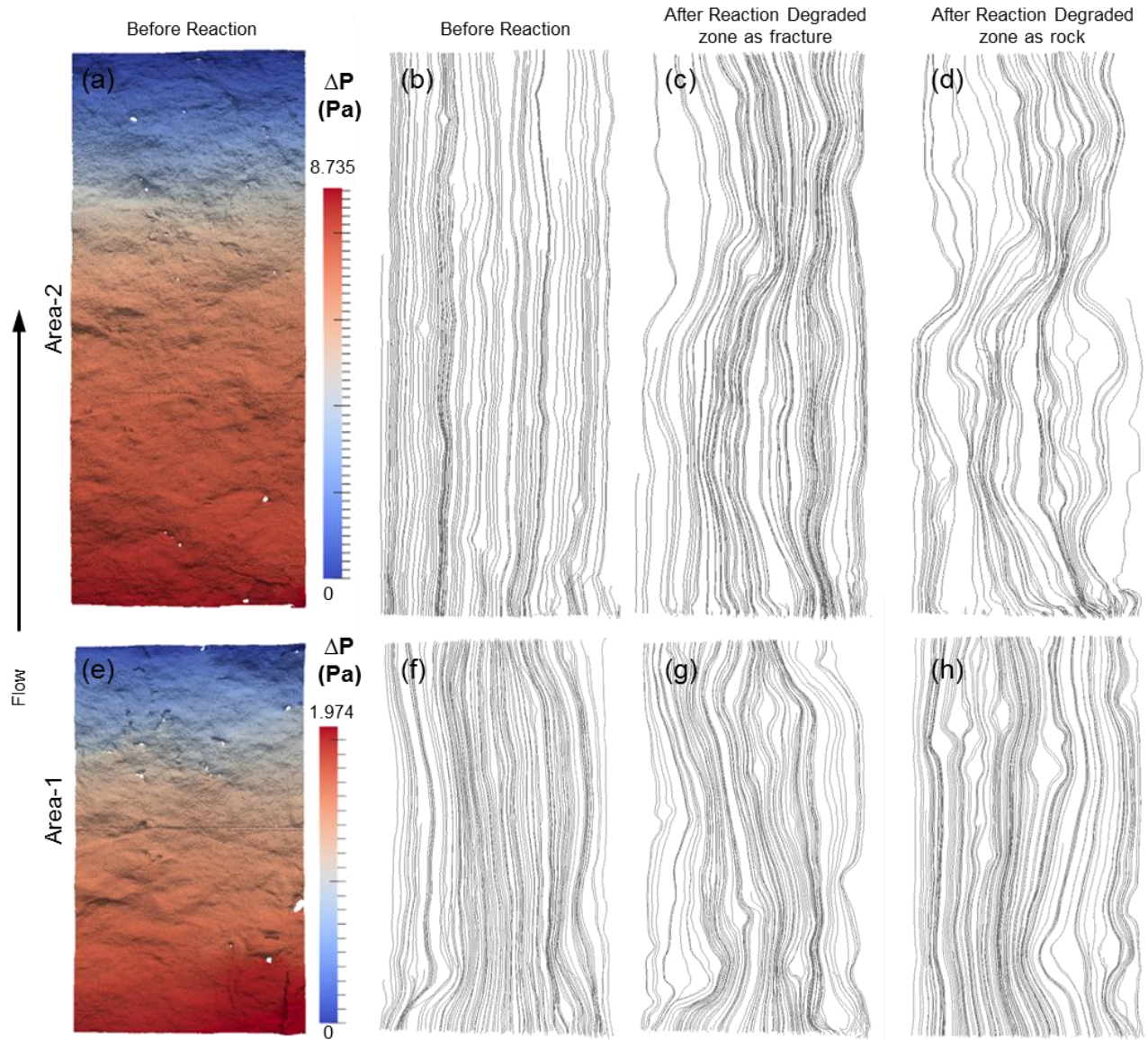


Figure 7. CFD simulation results: (a) Area-2 field of pressure difference (with respect to the outlet) in the before-reaction mesh (b)-(d) Area-2 velocity fields shown as streamlines for before-reaction mesh, after-reaction mesh with degraded zones treated as fracture and after-reaction mesh with degraded zones treated as rock, and (e)-(h) analogous for Area-1. Streamlines were generated by tracking the particles released at the inlet surface (at the bottom). In each geometry, 120 particles were released.

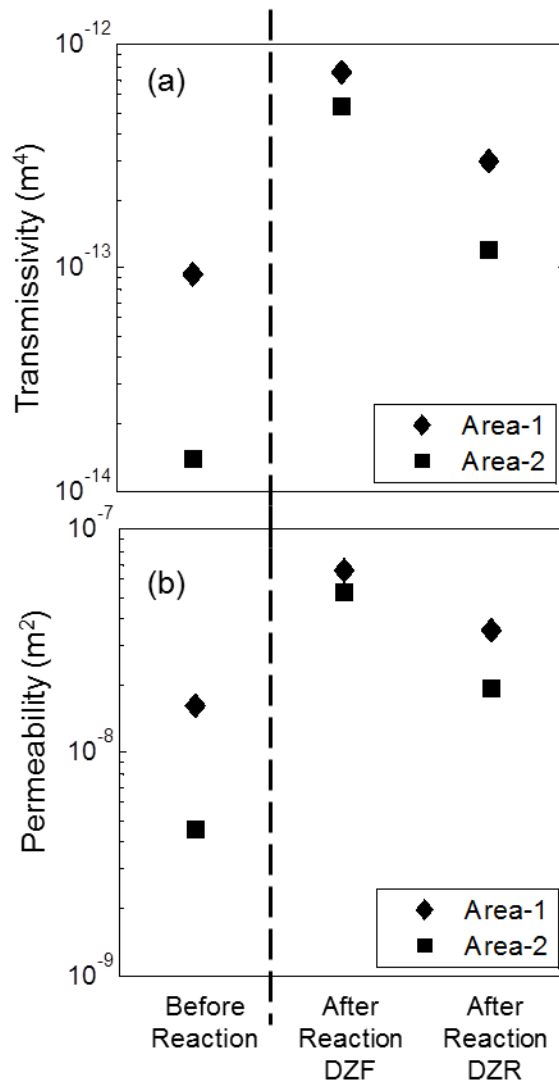


Figure 8. CFD simulation results of (a) transmissivity and (b) permeability of all six fracture geometries. DZF – Degraded Zones treated as Fracture, DZR – Degraded Zones treated as Rock

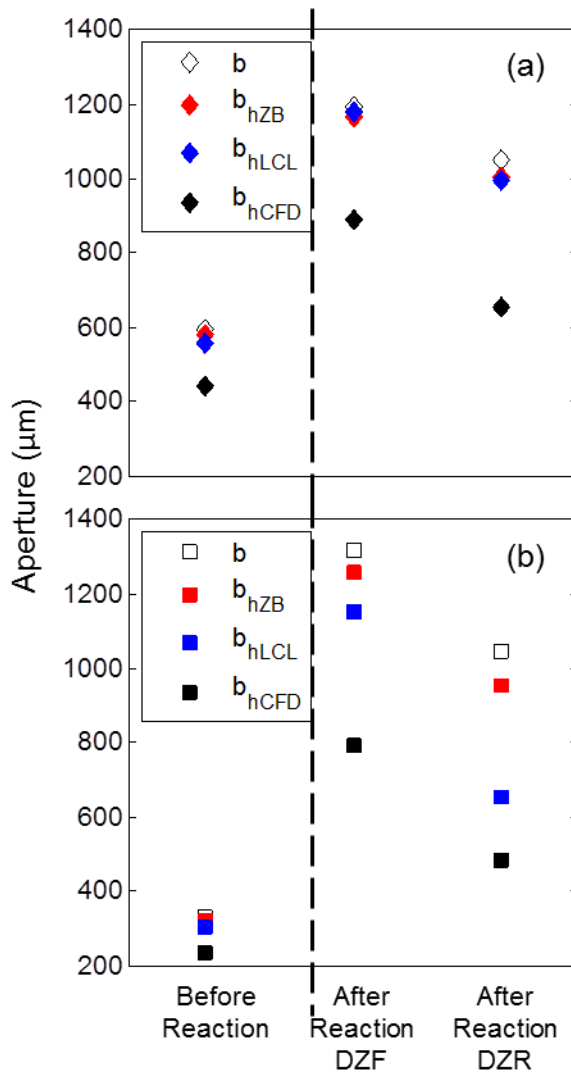


Figure 9. Mechanical aperture (b) and hydraulic apertures inferred from roughness model developed by Zimmerman and Bodvarsson (b_{hZB}), 2D Local Cubic Law (LCL) model (b_{hLCL}) and CFD simulations (b_{hCFD}) for (a) Area-1 and (b) Area-2. DZF – Degraded Zones treated as Fracture, DZR – Degraded Zones treated as Rock.

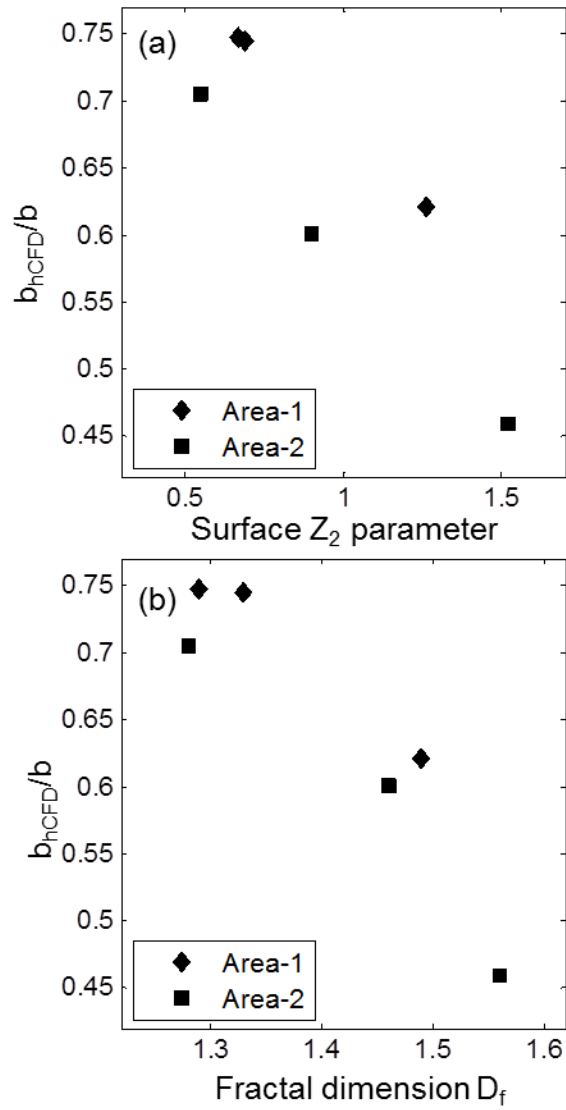


Figure 10. Ratio of hydraulic aperture inferred from CFD simulations over mechanical aperture (b_{hCFD}/b) in comparison with (a) surface Z_2 parameter and (b) fractal dimension D_f of the midsurface.

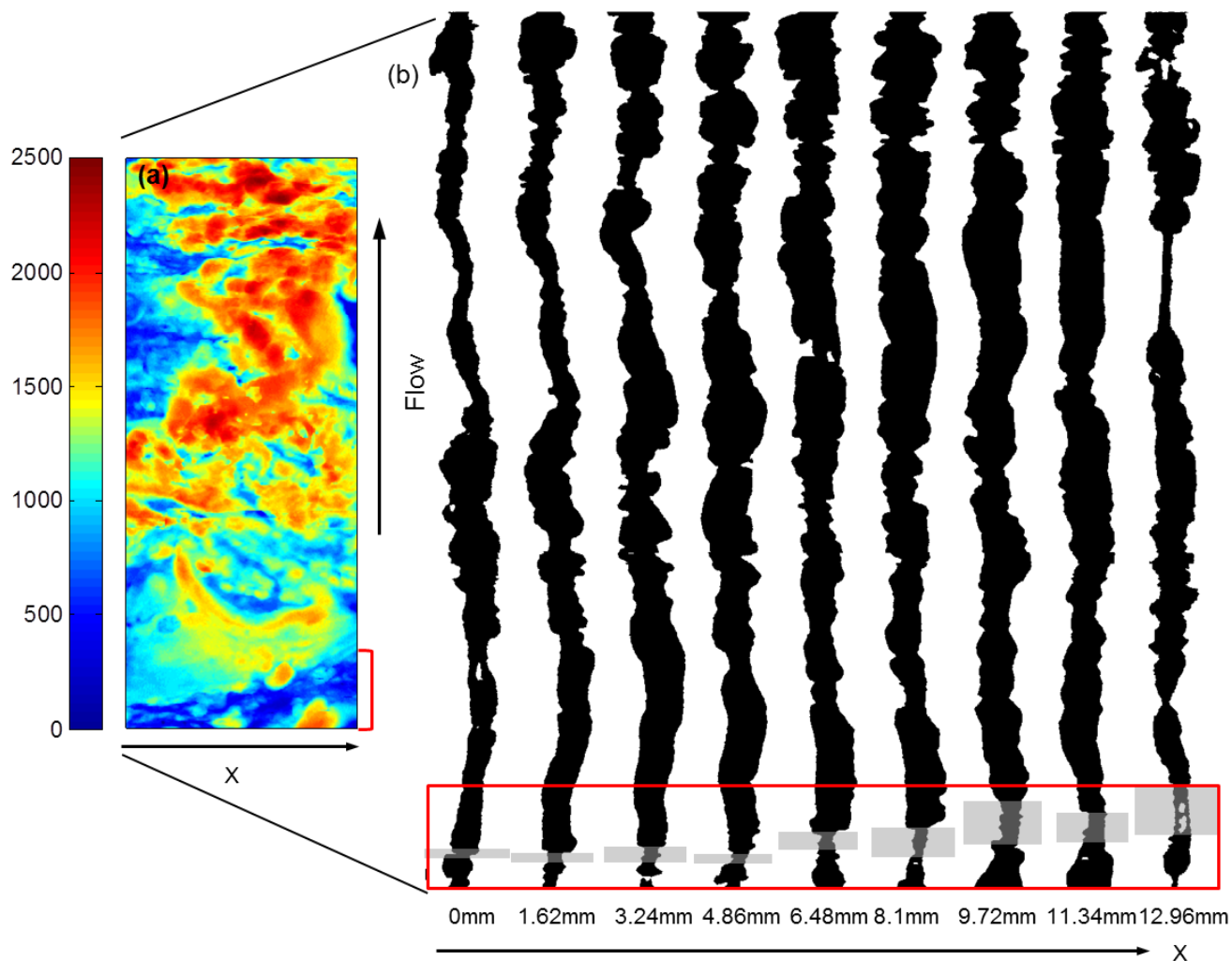


Figure 11. (a) Aperture map (μm) of after-reaction geometry of Area-2 with degraded zones treated as fracture, and (b) Sections of the fracture along the flow direction sampled every 1.62mm. Gray boxes highlight the areas where aperture increase is minimal, corresponding to the blue band near the bottom of the Area-2 fracture.

710

711 **Table 1.** Mesh properties of six fracture geometries. DZF - Degraded Zones treated as Fracture,
 712 DZR - Degraded Zones treated as Rock.

			Volume (mm ³)	# of grids	Surface Z ₂ parameter			<i>D_f</i>		
					Upper wall	Lower wall	Mid- surface	Upper wall	Lower wall	Mid- surface
Area-1	Before Reaction		1.68E+02	3.76E+06	0.85	0.97	0.67	1.33	1.36	1.29
	After Reaction	DZF	3.38E+02	6.41E+06	0.97	0.98	0.69	1.39	1.41	1.33
		DZR	2.98E+02	7.09E+06	1.66	1.92	1.26	1.51	1.56	1.49
Area-2	Before Reaction		1.33E+02	3.48E+06	0.66	0.79	0.55	1.31	1.33	1.28
	After Reaction	DZF	5.28E+02	5.23E+06	1.31	1.28	0.90	1.49	1.49	1.46
		DZR	4.21E+02	9.79E+06	2.17	2.27	1.52	1.57	1.57	1.56

713

714 **Table 2.** Aperture statistics of six fracture geometries. DZF – Degraded Zones treated as
 715 Fracture, DZR – Degraded Zones treated as Rock.

			Aperture (b) (μm)					
			min	max	mean	median	std	mode
Area-1	Before Reaction		0	1269	594	594	124	594
	After Reaction	DZF	0	1998	1194	1242	242	1323
		DZR	0	1998	1053	1080	303	1296
Area-2	Before Reaction		0	810	332	324	79	270
	After Reaction	DZF	0	2484	1319	1323	390	1377
		DZR	0	2457	1048	1053	420	1026

716